

Chapter 2, Problem 1P

Problem

Design a problem, complete with a solution, to help students to better understand Ohm's law. Use at least two resistors and one voltage source. Hint, you could use both resistors at once or one at a time, it is up to you. Be creative.

Step-by-step solution

Step 1 of 3

The one voltage source and two resistors circuit is shown in Figure 1.

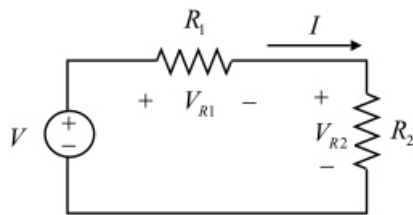


Figure 1

Step 2 of 3

The Ohm's law states that, the voltage across a resistor is directly proportional to the current flowing through the resistor.

Using Ohm's law, calculate the current through resistors and voltage across individual resistors in Figure 1.

The current through branch is,

$$I = \frac{V}{R_1 + R_2}$$

The voltage across the R_1 is,

$$V_{R1} = IR_1$$

Substitute $\frac{V}{R_1 + R_2}$ for I .

$$\begin{aligned} V_{R1} &= \left(\frac{V}{R_1 + R_2} \right) R_1 \\ &= V \left(\frac{R_1}{R_1 + R_2} \right) \end{aligned}$$

Therefore, the voltage across the R_1 is, $V \left(\frac{R_1}{R_1 + R_2} \right)$.

Step 3 of 3

The voltage across the R_2 is,

$$V_{R2} = IR_2$$

Substitute $\frac{V}{R_1 + R_2}$ for I .

$$\begin{aligned} V_{R2} &= \left(\frac{V}{R_1 + R_2} \right) R_2 \\ &= V \left(\frac{R_2}{R_1 + R_2} \right) \end{aligned}$$

Therefore, the voltage across the R_2 is, $V \left(\frac{R_2}{R_1 + R_2} \right)$.